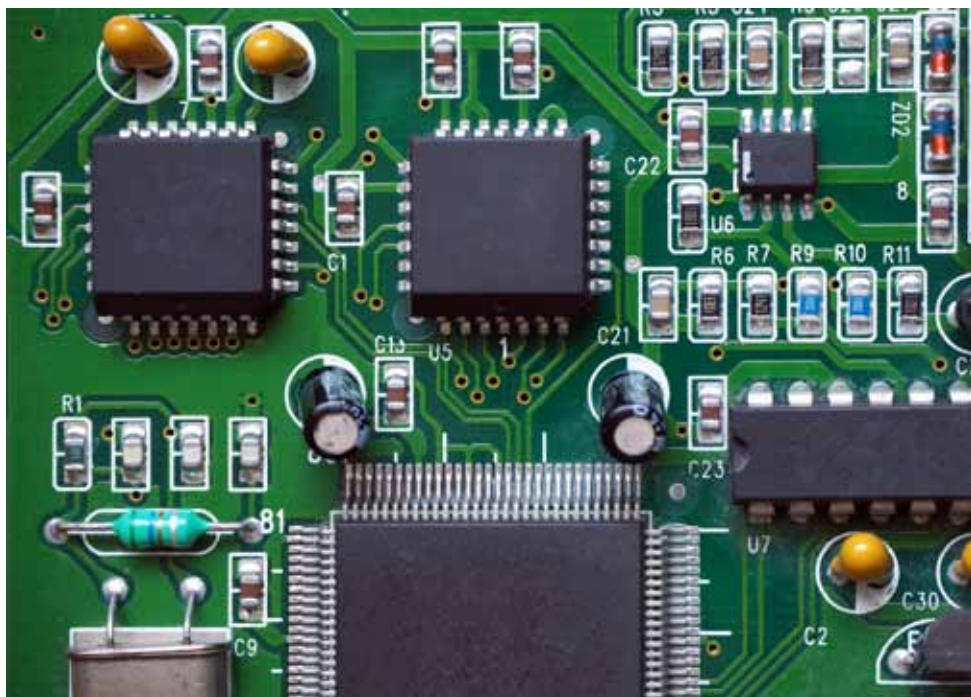


# CHIPLETS Demystified

The semiconductor industry is betting on this emerging technology to hoodwink Moore's Law. Here are the basic details everyone should know about chiplets.



**JANANI G. VIKRAM** is a freelance writer based in Chennai, who loves to write on emerging technologies and Indian culture.



## What are chiplets?

- Chiplets are small, modular integrated circuits (ICs) with limited functions or features.
- Traditional monolithic design packs all the components into a single large piece of silicon. Chiplets separate the functionality into specialised dies, which are then assembled into a single package.
- Basically, take a system-on-chip (SoC) and break it down into its functional components—general-purpose processing, custom accelerators, data storage, high-speed input and output (I/O) communications. Now, make separate chips specialising in each of these tasks. You now get CPU chiplets, GPU chiplets, memory chiplets, and I/O

chiplets, which you can put together like 'Lego' building blocks!

- Now what happens is that each component becomes a separate chip – which can scale to the limit of Moore's Law. Collectively, they can surpass the limitation!

## Why chiplets?

- **Reusable.** The same chiplets can be reused in various devices. This increases flexibility and reduces costs and time-to-market.
- **Known good die.** Since the chiplets are already tested and guaranteed to be error-free by their manufacturers, the yield and reliability of the final assembly are higher.
- **Heterogeneous integration.** You can